

Maths Adventure for National Science Week: The Lost Treasure of Coral Bay

Year 4 · National Science Week Special

ACARA V9.0 — AC9M4N06, AC9M4N07, AC9M4N08 (Number: efficient calculation strategies, estimation and rounding, mathematical modelling)

Ahoy, Maths Explorer! Captain Ruby's treasure map was torn apart in a wild storm off the coast of Coral Bay! She has found four scattered map pieces — but each one is locked behind a maths clue. Solve every clue correctly, crack the final code, and help Captain Ruby find the lost treasure before the tide comes in!

Learning Intention: Students will solve addition, subtraction, multiplication, and division word problems using multi-digit numbers, and use their answers to crack a numerical code.

Warm-Up: Ready Your Sea Legs

$234 + 156 = \underline{\quad}$

$500 - 267 = \underline{\quad}$

$7 \times 8 = \underline{\quad}$

$144 \div 12 = \underline{\quad}$

The Adventure: Solve the Map Clues

CLUE 1 · ADDITION

Captain Ruby found 246 gold coins in one chest and 178 gold coins in another. How many gold coins does she have altogether?

Answer:

CLUE 2 · ADDITION

The ship sailed 312 km on Monday and 275 km on Tuesday. How far did it sail in total?

Answer: km

CLUE 3 · SUBTRACTION (MONEY)

Ruby had \$150 to spend on supplies. She spent \$68. How much money does she have left?

Answer:

CLUE 4 · MULTIPLICATION

There are 7 pirate ships in Ruby's fleet, each carrying 8 crew members. How many crew members are there in total?

Answer:

CLUE 5 · DIVISION

A treasure of 144 gold coins is shared equally among 12 pirates. How many gold coins does each pirate get?

Answer:

CLUE 6 · TIME

Ruby left the ship at 8:45am. The journey to the treasure took 1 hour and 35 minutes. What time did she arrive?

Answer:

Unlock the Treasure Chest

Take the **ones digit** from your answers to Clues 1, 2, 3, and 4. Write them in order to reveal the 4-digit lock code!

Clue 1 → Clue 2 → Clue 3 → Clue 4

Challenge: Estimate the Treasure

Ruby estimated she would need about 300 gold coins for the journey. She actually has 287 coins. Round 287 to the nearest ten, then to the nearest hundred. Was her estimate reasonable?

Nearest ten:

Nearest hundred:

Bonus Challenge: Sharing with Leftovers

250 gold coins are shared equally among 8 pirates. How many gold coins does each pirate get, and how many coins are left over?

Answer:

Design Your Own Pirate Coin

You found the treasure! Design your own gold coin below and give it a name.

Case Status

Clues solved correctly: ____ / 6



Treasure
Found!



So Close — Try
Again

Parent/Teacher Tip: Encourage students to estimate an answer before calculating (e.g. 'about 400 + 200') so they can immediately spot a calculation error if their final answer is way off the estimate.

For teachers and parents: use this page to check student responses.

Warm-Up Answers

$234 + 156 = 390$ $500 - 267 = 233$ $7 \times 8 = 56$ $144 \div 12 = 12$

Adventure Clue Answers

Clue 1: $246 + 178 = 424$ gold coins

Clue 2: $312 + 275 = 587$ km

Clue 3: $\$150 - \$68 = \$82$

Clue 4: $7 \times 8 = 56$ crew members

Clue 5: $144 \div 12 = 12$ gold coins each

Clue 6: $8:45\text{am} + 1 \text{ hour } 35 \text{ minutes} = 10:20\text{am}$

Treasure Chest Code

Ones digit of Clue 1 (424) → **4** Clue 2 (587) → **7** Clue 3 (82) → **2** Clue 4 (56) → **6**

Final Code: 4 - 7 - 2 - 6

Challenge & Bonus Answers

Estimate the Treasure: 287 rounds to **290** (nearest ten) and **300** (nearest hundred) — Ruby's estimate of 300 was reasonable.

Sharing with Leftovers: $250 \div 8 = 31$ coins each, with **2** coins left over.